DSAN 6725 · FINAL PROJECT

AN AGENTIC AI SYSTEM

Sage

A grounded, memory-aware meal planner that sits between manual trackers and free-generating LLMs.

Tyler Blue · Andrew Moy

FINAL PROJECT PRESENTATION

DSAN 6725 · SPRING 2026

APRIL 24, 2026

The current state of meal planning tools

Most digital meal-planning tool falls into one of two categories

MANUAL TRACKING

01

High cognitive overhead

Every meal, every ingredient, logged by hand. Accurate in principle — but long-term adherence collapses. No memory of the user between sessions.

MYFITNESSPAL | MEALIME | EATTHISMUCH

FREE-FORM LLM

02

Low friction, low reliability

Fluent natural language, zero tracking burden — but hallucinated meals, invented macros, and no persistent context about who the user is.

CHATGPT DIY | GENERIC AI APPS

→ What's missing: a system that feels conversational but stays grounded in real nutritional data.

THE THESIS

A system that feels like talking to someone who already knows your goals — while having access to the data to actually meet them.

01

Multi-agent pipeline

Orchestrator + Intake + Planning agents with narrow, well-defined responsibilities.

02

Persistent memory via Mem0

User profile survives across sessions. The system remembers goals, restrictions, and preferences.

03

Grounded retrieval + audit

Meals come from a curated 751-recipe corpus. A deterministic audit layer validates every plan.

FROM RELATED WORK

~35%

Five leading LLMs undercalculated daily energy intake by ~35% (≈ 695 kcal) versus registered-dietitian reference plans.

BILEN ET AL. · 2026 · N = 5 MODELS

23 studies

A systematic review flagged safety concerns, hallucination risks, and limited real-world implementation across the ChatGPT meal planning literature.

GUO ET AL. · 2025 · SYSTEMATIC REVIEW

DESIGN IMPLICATION

Don't let the LLM invent meals. Have it assemble from a corpus with pre-validated nutrition.

SOURCE · BUDGET BYTES

Recipe Database

Selected for transparent per-serving cost data. Scraped via sitemap traversal; structured recipe fields pulled from WordPress Recipe Maker markup.

751
RECIPES
After cleaning & dedup

$1.22
MEDIAN COST / SERVING
Budget-friendly by design

35 min
MEDIAN TOTAL TIME
Prep + cook

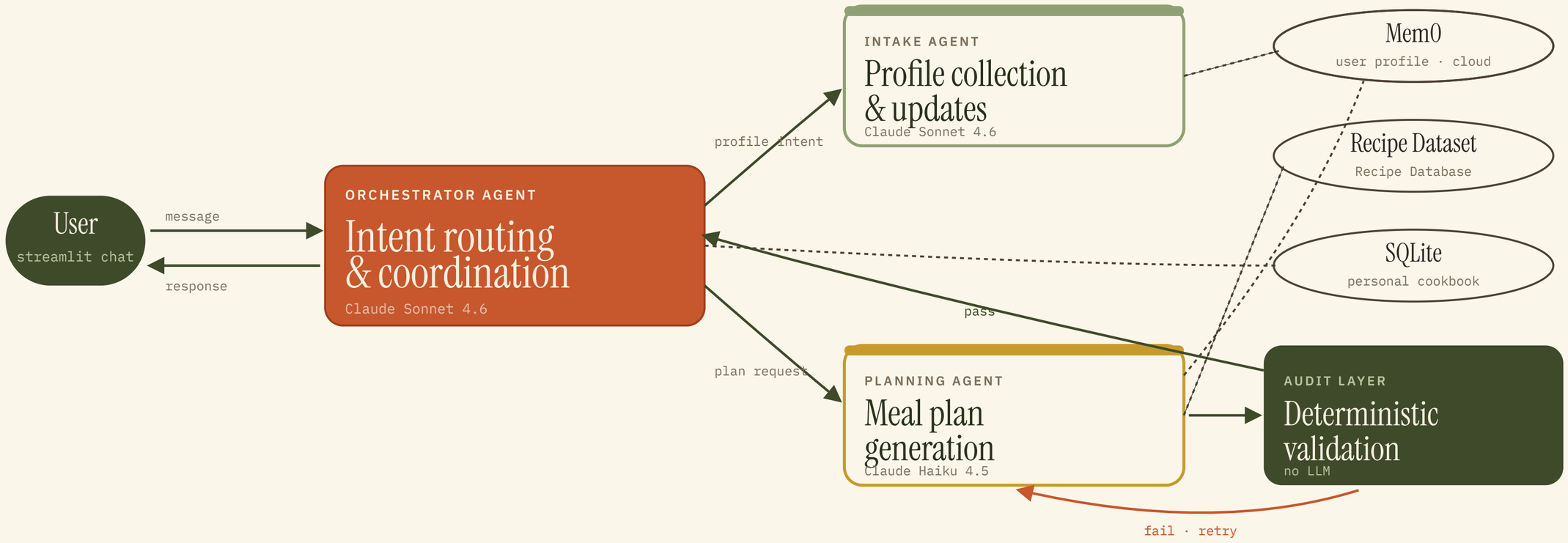
213 kcal
MEAN / SERVING
Range: < 50 – 675 kcal

Nutrition values are LLM-estimated, not sourced from USDA FoodData Central. A deliberate trade-off to enable rapid pipeline development. For production, the corpus would be replaced with a lab-validated dataset.

THE BIG PICTURE

System Architecture

STRANDS SDK · MEM0 · SQLITE



SPECIALIZATION > GENERALIZATION

Agent Breakdown

01 · USER-FACING

Orchestrator

Claude Sonnet 4.6

Analyzes each message, routes intent to the right sub-agent, handles cookbook CRUD directly.

TOOLS

- update_user_profile
- build_meal_plan
- audit_meal_plan
- generate_shopping_list
- add_recipe_to_cookbook
- add_recipe_from_url
- edit_cookbook_recipe

02 · PROFILE

Intake Agent

Claude Sonnet 4.6

Builds and updates the user's profile through natural conversation — one or two questions at a time.

TOOLS

- get_user_profile
- update_goal
- update_body_stats
- update_activity_level
- update_dietary_preferences
- update_meal_constraints
- recall_user_info

03 · PLAN GENERATION

Planning Agent

Claude Haiku 4.5

Computes targets deterministically (Mifflin-St Jeor), queries the corpus, assembles scored day or week plans.

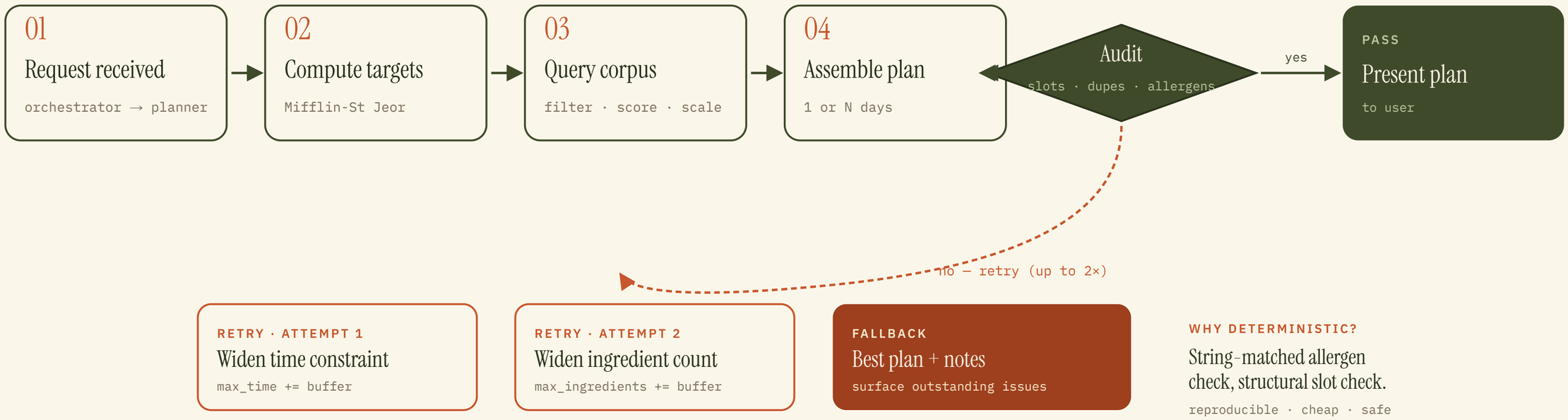
TOOLS

- get_user_profile
- estimate_calorie_target
- estimate_protein_target
- build_day_plan_tool
- build_week_plan_tool
- search_meals_tool

OUR SAFETY MECHANISM

Plan-Audit Loop

No LLM in the validation path. Every plan is checked for required slots, duplicates, and allergen violations before it reaches the user.



WHY DETERMINISTIC?

String-matched allergen check, structural slot check.

reproducible · cheap · safe

EVERY CHOICE MINIMIZES ACCIDENTAL COMPLEXITY

Architecture Decisions

Strands Agent SDK

FRAMEWORK

Transparent tool-calling — no chains, runnables, or callbacks hiding what the model is doing.

chosen over · LangChain

Claude Sonnet 4.6

REASONING AGENTS

Open-ended reasoning for intent routing and ambiguous intake conversation.

used by · orchestrator, intake

Claude Haiku 4.5

STRUCTURED AGENT

Sequential tool-calls on a fixed workflow — faster and cheaper than Sonnet.

used by · planner

Mem0

MEMORY · PROFILES

Cloud vector memory keyed by user ID. Handles embeddings internally — no infra to manage.

chosen over · local vector DB

SQLite

COOKBOOK

Relational data — custom recipes and exclusions. No server, works identically local or deployed.

not a vector store job

Streamlit

FRONTEND

Chat-style UI with minimal frontend overhead — forwards messages, renders responses.

chosen over · custom React

BRAINTRUST HARNESS · ALL DETERMINISTIC SCORERS

29 cases, 8 scorers

TEST CASES BY CATEGORY

Meal Planning	12
Dietary Restriction	6
Profile Update	4
Out of Scope	4
Shopping List	3
Total	**29**

SCORERS · ALL DETERMINISTIC

`ToolSelection`
Recall of expected tools

`Latency`
Banded response time

`NoError`
No exceptions thrown

`ScopeAwareness`
In-scope / refusal

`PlanCompleteness`
Required plan fields

`CalorieTarget`
Within ±15%

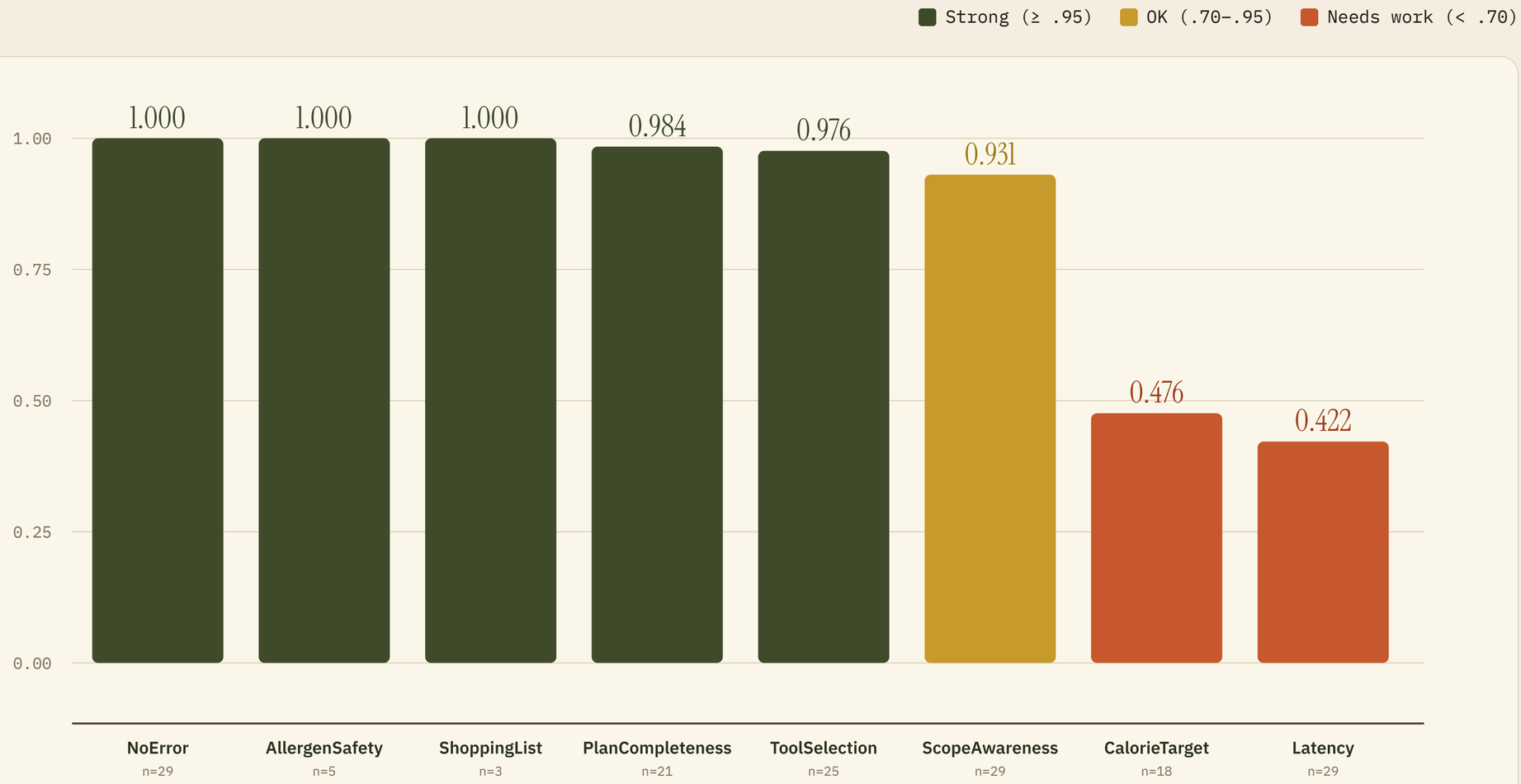
`AllergenSafety`
No restricted ingredients

`ShoppingList`
Quantities + line items

Why no LLM-as-judge? Reproducible, cheap, and auditable.

AVERAGE SCORE ACROSS APPLICABLE CASES

All Evaluations



THREE LOWER SCORES — AND WHAT WE'D FIX

Where the system struggles

.476

CALORIETARGET

WHY

Budget Bytes skews light — median 213 kcal/serving. Muscle-gain profiles can't reach targets even with 1.5× serving multipliers.

FIX

Replace corpus with a lab-validated, macro-diverse dataset (e.g., USDA-backed). The pipeline is corpus-agnostic.

.422

LATENCY

WHY

2–3 sequential Claude API calls per turn, plus Mem0 reads. Multi-day plans fall into the 60–120s band.

FIX

Parallelize independent tool calls, cache Mem0 reads per session, move to streaming responses so perceived latency drops.

.931

SCOPEAWARENESS

WHY

Two out-of-scope cases scored 0. The system responded thoughtfully and safely — it just didn't use the refusal phrase the scorer expects.

FIX

Scorer calibration, not a system change. We'd either tune the scorer's regex or add a lightweight refusal classifier.

SWITCHING TO STREAMLIT

Demo

Profile intake · Multi-day plan · Allergen case · Shopping list

01 · ONBOARD A NEW USER
02 · BUILD A 3-DAY PLAN
03 · TRIGGER AN ALLERGEN RETRY
04 · GENERATE SHOPPING LIST

WHAT WE SHOWED

Conversational Multi-Agent Meal Planning

An agentic architecture can deliver conversational UX *and* general nutritional reliability.

DSAN 6725 · APRIL 2026

Slides crafted with Claude Cowork

FUTURE WORK

Lab-validated corpus

USDA FoodData Central — fixes CalorieTarget

Grocery MCP integration

Plan → cart, end-to-end

Temporal profile

Weight & intake tracking; adaptive targets

Production deploy

Auth, hosting, polished UI

Tyler Blue · Andrew Moy

THANK YOU · QUESTIONS?